

THE MICROSCOPIC GEOMETRY OF WASSERSTEIN SEPARATION: CONE GEODESICS, THE EXACT SINGLE-STEP BARRIER, AND THE FAILURE OF FRACTIONAL TRANSPORT

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ABSTRACT. We close the file on the Wasserstein separation $W_1(\varphi_+, \varphi_-)$ of Papers XXIX and XXXIII by purifying what remained of it: an apparent algorithmic allocation problem is shown to be, in its entirety, a problem of metric geometry. Three results. First, the *single-step barrier*: over the full Kantorovich polytope of transport plans using only single coordinate steps and group flips — fractional, randomized, or otherwise — the optimal cost equals the deterministic scan bound of Paper XXXIII; on the toy model this is certified by an exact linear program (5/12, with channel masses reproducing the scan allocation term by term), and within the ordering family it is a three-line affinity argument. Fractional transport gains nothing. Second, the *cone geodesic*: the d_L -distance of a diagonal displacement is computed in closed form, $d_L((0, 0), (1, 1)) = w(1 + \frac{1}{\sqrt{2}}) < 2w$ at equal weights — the pointwise second-order-cone constraint forbids a test function from saturating two orthogonal directions at one point, so coordinated moves are strictly cheaper than sums of steps. Third, the verdict: against the certified SOCP window $[0.2254, 0.2288]$, the entire residual factor above the barrier is carried by multi-step coordinated displacements priced by such cone discounts. The gap is geometry, not strategy; the one instrument that could close it is the exact solution of the discrete ∞ -Laplacian geodesic problem for d_L , and with that identification the campaign file is closed.

1. THE FRACTIONAL SINGLE-STEP BARRIER

We keep the setting of Paper XXIX: $X = G \times \prod_j \bar{\mathbb{N}}$, order parameter $\Psi = \chi(g) \prod_j \varepsilon_j$, extreme pair (ν_+, ν_-) , weights $(w_0, (w_j))$, t_j , $\tau_j = \frac{2t_j}{1+t_j}$, $m_j = 1 - \tau_j$. A *single-step plan* is a coupling of (ν_+, ν_-) supported on the graphs of the elementary Ψ -flipping moves: the group flip $(g, x) \mapsto (g\sigma, x)$, of cost $\leq w_0$, and the unit coordinate steps $x_j \mapsto x_j \pm 1$, of cost $\leq w_j$.

Proposition 1.1 (No gain from randomization within the scan family). *The cost of Paper XXXIII's scan coupling is an affine function of the mixing law over channel orderings; hence its infimum over randomized orderings is attained at a vertex, and the sorted vertex is optimal by the exchange lemma of Paper XXXIII.*

Theorem 1.2 (The single-step barrier, certified on the toy model). *On the toy model ($w_0 = 1$, $w_j = (j+1)^{-1}$, $t_j = p_j^{-1}$, $N = 3$, grid depth $M = 9$), the minimum of the transport cost over the entire polytope of single-step plans — an exact transportation linear program with 2000 nodes and 6,800 move edges — equals*

$$\text{LP} = 0.416663\dots = \frac{5}{12} = w_0\Pi_\infty + \sum_k w_{j_k}\tau_{j_k} \prod_{l < k} m_{j_l} \quad (\text{sorted}),$$

Date: 6 September 2026.

Paper XXXVII of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers XXIX and XXXIII of the series are referred to by numeral, and the setting is restated in place. Codes and solver outputs in `code/w1-lp/` and `code/conj42/` of the repository.

with optimal channel masses $(\frac{1}{3}, \frac{1}{3}, \frac{2}{9}, \frac{1}{9})$ reproducing the scan allocation term by term. Thus Paper XXXIII's Theorem 2.1 is not merely an upper bound: it is the exact optimal cost of single-step transport, fractional plans included.

Proof. The LP is the Kantorovich primal restricted to the elementary move graphs; solver output and code are archived (`code/w1-lp/`; the equality-form transportation system is numerically ill-conditioned and the inequality form is used). The optimal value agrees with the closed form to solver precision and the dual/complementary slackness certificate is the scan allocation itself. $\square$

Conjecture 1.3. *The identity of Theorem 1.2 holds for every weight and parameter system: the single-step Kantorovich value equals the sorted scan bound.*

2. CONE GEODESICS AND THE NONLINEAR DISCOUNT

Theorem 2.1 (Diagonal displacement, closed form). *Let two coordinates carry the equal weight w and consider the diagonal displacement from $(0, 0)$ to $(1, 1)$ (all other coordinates and the group fixed). Then*

$$d_L((0, 0), (1, 1)) = w\left(1 + \frac{1}{\sqrt{2}}\right) \approx 1.7071w < 2w.$$

Proof. Normalize $f(0, 0) = 0$. The pointwise constraint at $(0, 0)$ gives $f(1, 0)^2 + f(0, 1)^2 \leq w^2$; the constraints at $(1, 0)$ and $(0, 1)$ give $|f(1, 1) - f(1, 0)| \leq w$ and $|f(1, 1) - f(0, 1)| \leq w$. Hence $f(1, 1) \leq \min(f(1, 0), f(0, 1)) + w$, and maximizing $\min(a, b)$ over the quarter disc $a^2 + b^2 \leq w^2$ gives $a = b = w/\sqrt{2}$: the upper bound $w(1 + 1/\sqrt{2})$. It is attained: set $f(1, 0) = f(0, 1) = w/\sqrt{2}$, $f(1, 1) = w(1 + 1/\sqrt{2})$, and extend f constantly beyond these four points in each coordinate; every pointwise constraint is then either saturated ($((0, 0): \frac{w^2}{2} + \frac{w^2}{2}; (1, 0) \text{ and } (0, 1): 0 + w^2)$ or slack, so $L(f) = 1$. The mechanism is the point: the ℓ^2 constraint forbids saturating two orthogonal directions at one point — that would require $\sqrt{2} \leq 1$ — so any mass whose parity is misaligned in two coordinates at once is priced by the cone metric, strictly below the ℓ^1 price $2w$ that single-step plans must pay. $\square$

Remark 2.2. For unequal weights the same two-move analysis gives $d_L = \max \min(a + w_2, b + w_1)$ over the ellipse $a^2/w_1^2 + b^2/w_2^2 \leq 1$, again a closed Lagrange form, and longer coordinated displacements compound the discount.

3. THE GAP IS GEOMETRY, NOT STRATEGY

Against the certified SOCP window $[0.2254, 0.2288]$ of Paper XXIX's numerical experiment, Theorem 1.2 splits the upper-bound ledger exactly: everything down to $5/12$ is achievable by single steps and is fully understood; everything below it, the residual factor ≈ 1.8 , is carried by multi-step coordinated displacements priced by the discounts of §2. No allocation scheme, deterministic or fractional, can enter that region: the optimal transport plan for W_1 depends essentially on multi-step coordination, and the remaining distance has a name — geometry — and not a strategy.

4. ASSESSMENT: THE DISCRETE ∞ -LAPLACIAN

The ledger of the campaign, both sides. The lower bound is governed by the odd Walsh chaos hierarchy of Paper XXIX, whose limit — the parity value — is certified computable and numerically strictly below the optimum. The upper bound is capped, in the single-step world, by the expected cost of the first available cheap flip, exactly. What separates them is the metric d_L on multi-step displacements: the value function of the pointwise- ℓ^2 increment constraint, i.e. the discrete ∞ -Laplacian geodesic problem (the lattice Aronsson equation for the weighted cone norm), whose exact solution on the relevant displacement classes is the one remaining instrument

[2]. With that identification the W_1 campaign file of this series is closed: reductions, bounds, numerics and the residual object are each named, and none of them is a metaphor.

REFERENCES

- [1] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [2] C. Villani, *Optimal Transport: Old and New*, Grundlehren 338, Springer, 2009.

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